

Qwen3 taxlili

Sardor Abdirayimov, 20 Iyun 2026

Muzyorar

O'ngga va chapga qarang, chatga yozing chatdan

Izmingiz nima?

Kasbingiz?

Hobby?

Bugun nima o'rgandingiz?

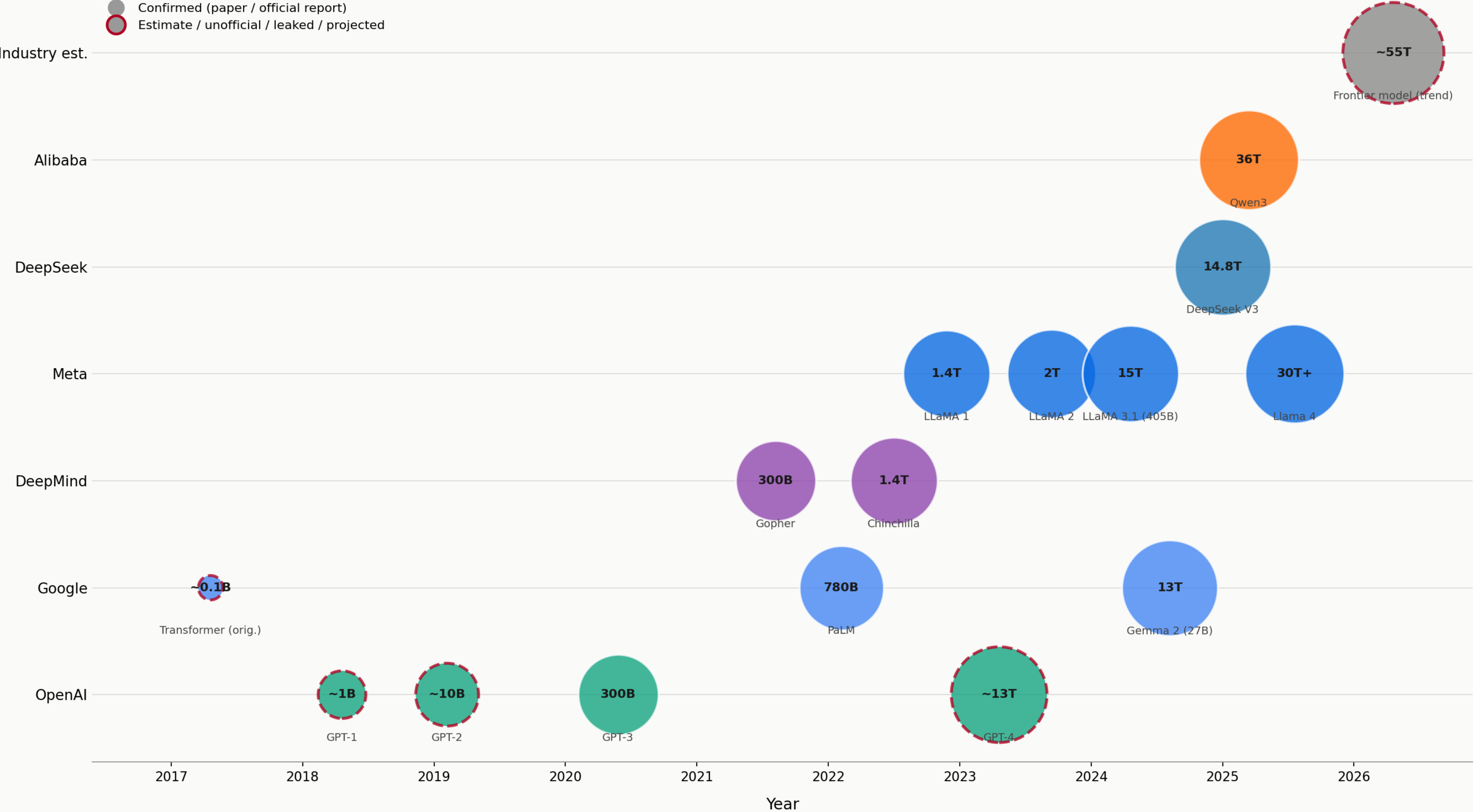
Men xaqimda

- Sardor Abdirayimov
- Xozir UNIST Instituteda Magistr qilyapman.
- RL va Generative AI ga qiziqaman.

LLM Technical Reports

160+ reports from 30+ labs | 2017-2026

Training Tokens Used by Major LLMs (2017-2026)
Bubble area scaled by $\log_{10}(\text{tokens})$ — spans ~0.1B to ~55T tokens



Sources: original model papers/technical reports (OpenAI, Meta, Google, DeepMind, DeepSeek, Alibaba); leaked/estimated figures noted with dashed red outline. Token counts for 2026 closed frontier models are not publicly disclosed; final bubble is a trend extrapolation, not a confirmed figure.

Tokenizer

Turi	Afzallik	Kamchilik	Misol
Word	Tushunishga oson	Juda katta vocabulary size	ayiq, daraxt
Subword	Soʻzlar ildizlarga boʻlinishadi	tokenization qiyinlashadi	ay-iq, dar-a-xt
Character, Byte	Kichik vocabulari size	Juda uzun “sequence length”	a-y-i-q, d-a-r-a-x-t

Byte Pair Encoding algoritmi asosan ishlatiladi

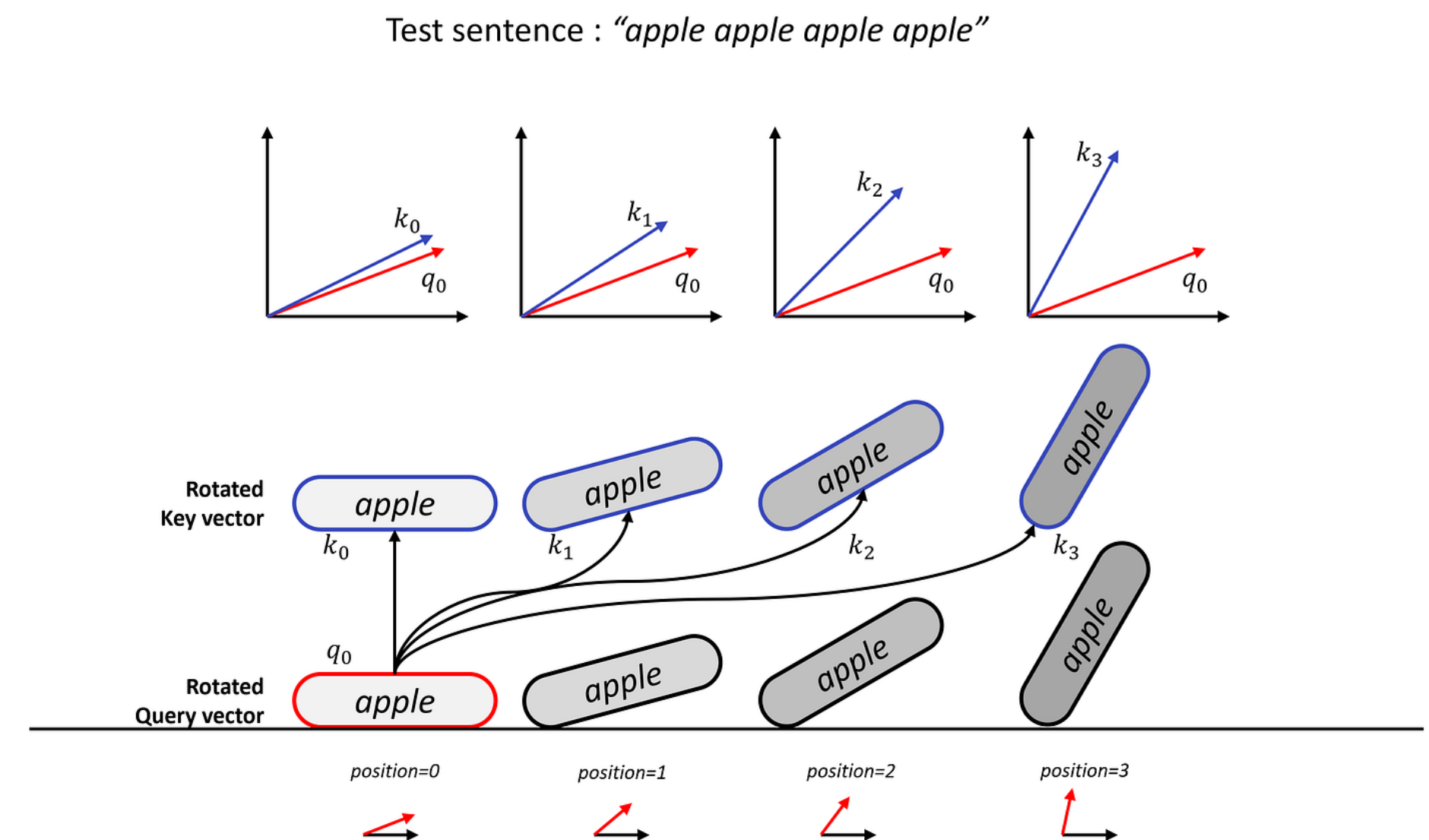
Embedding

Qozonga yursang qorasi yuqadi
Yuqadi qorasi yursang qozonga

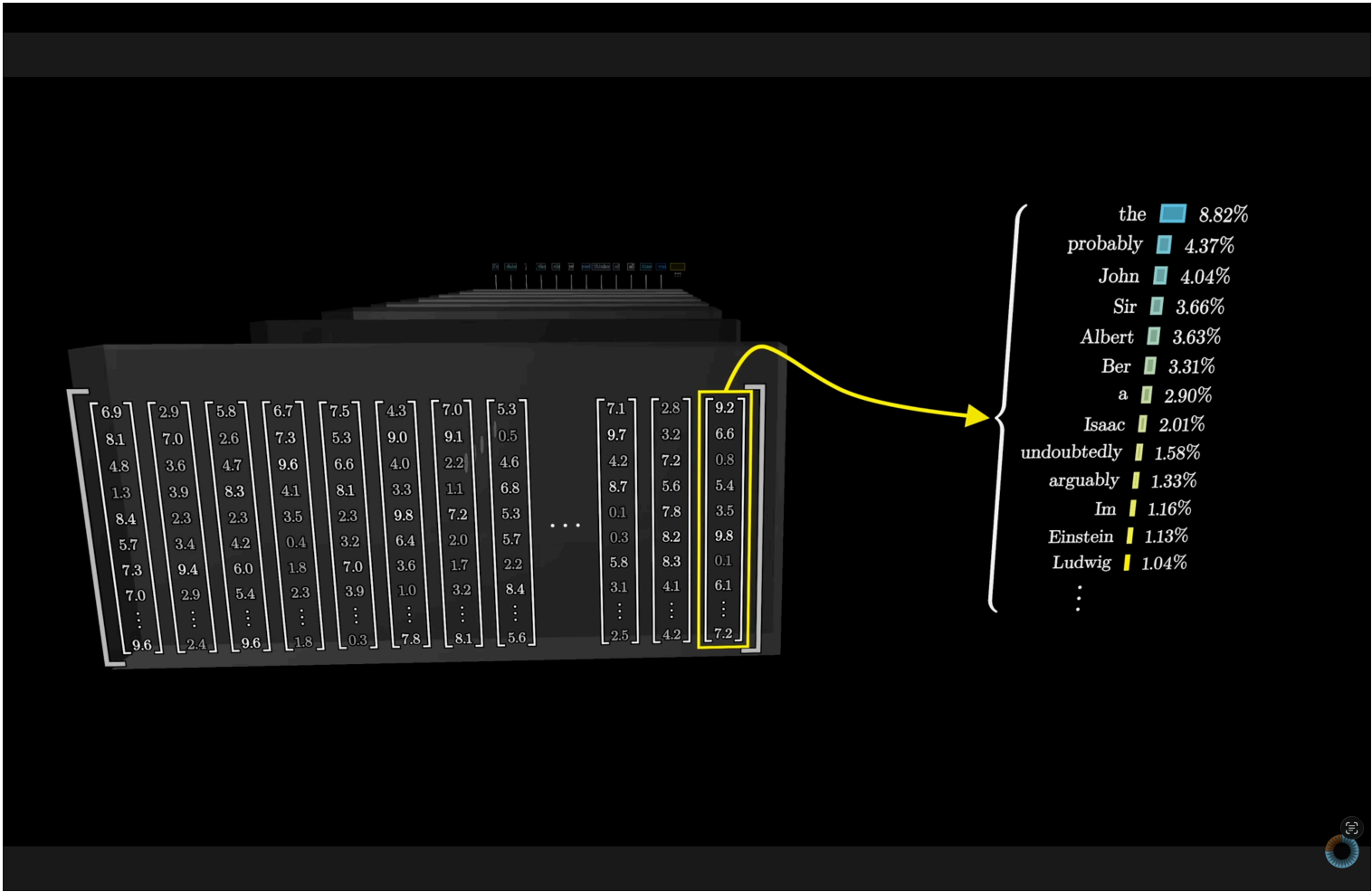
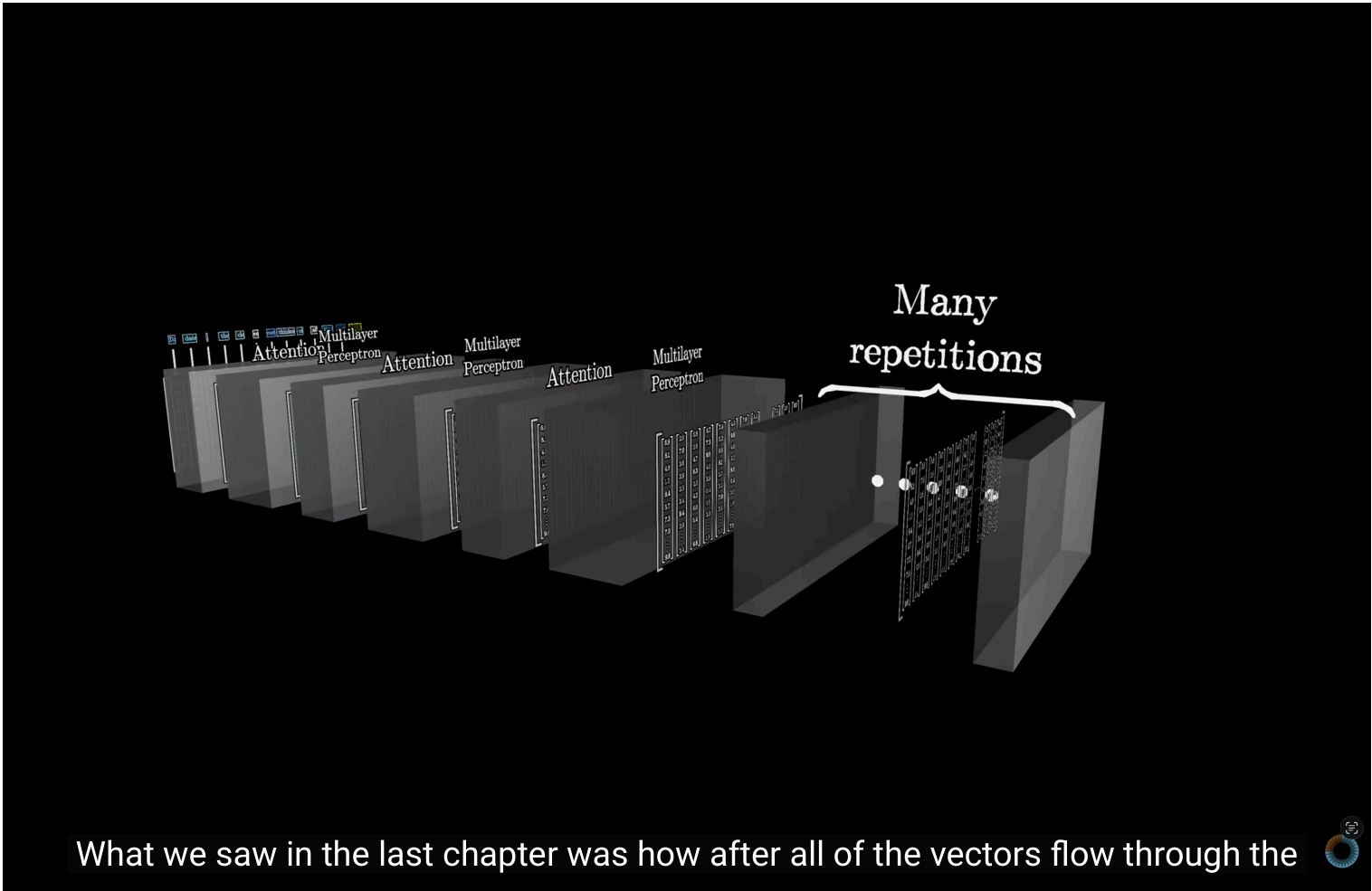
Positional Encoding

Periodic Positional Encoding

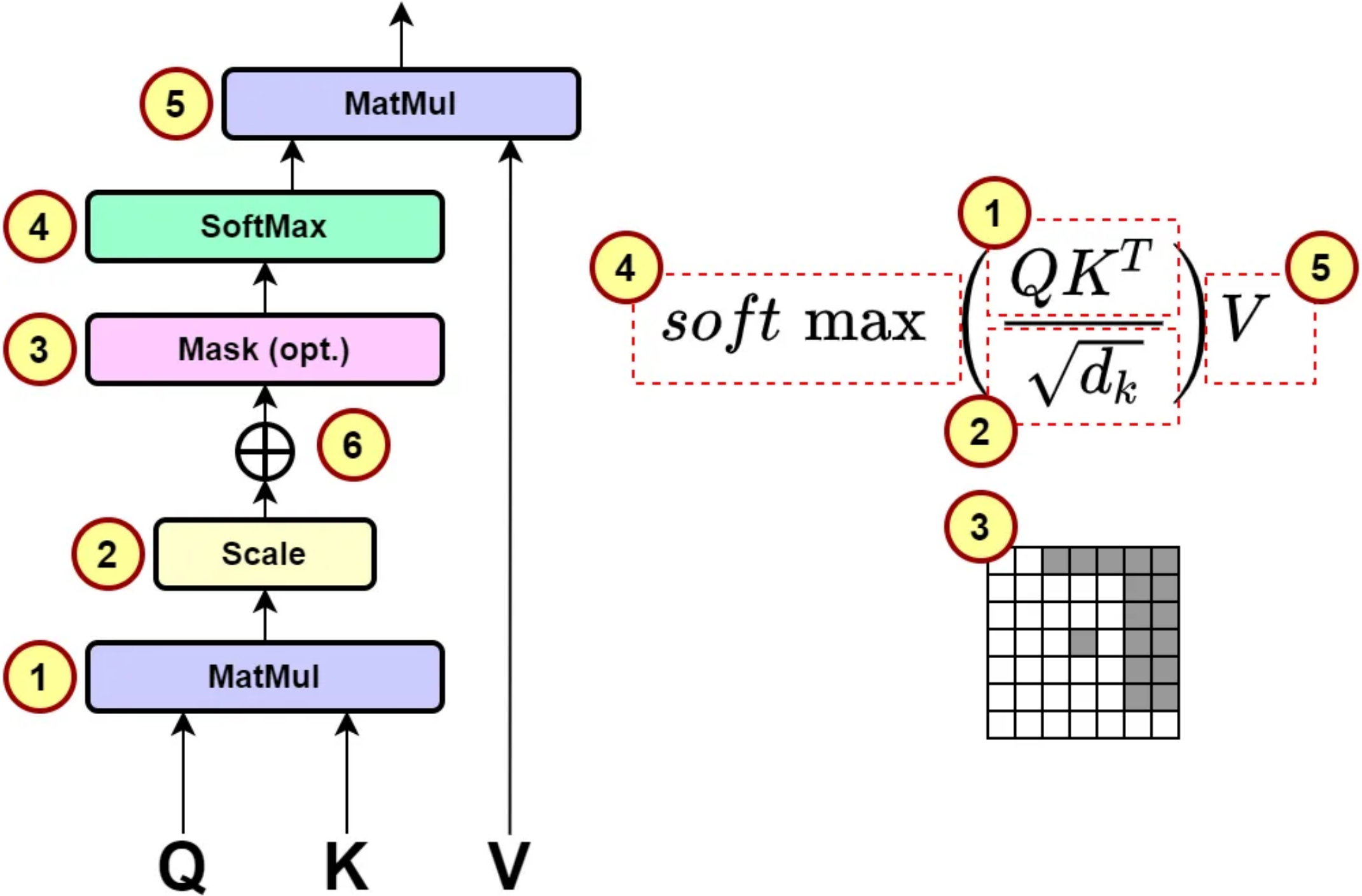
Rotational Positional Encoding (RoPE)



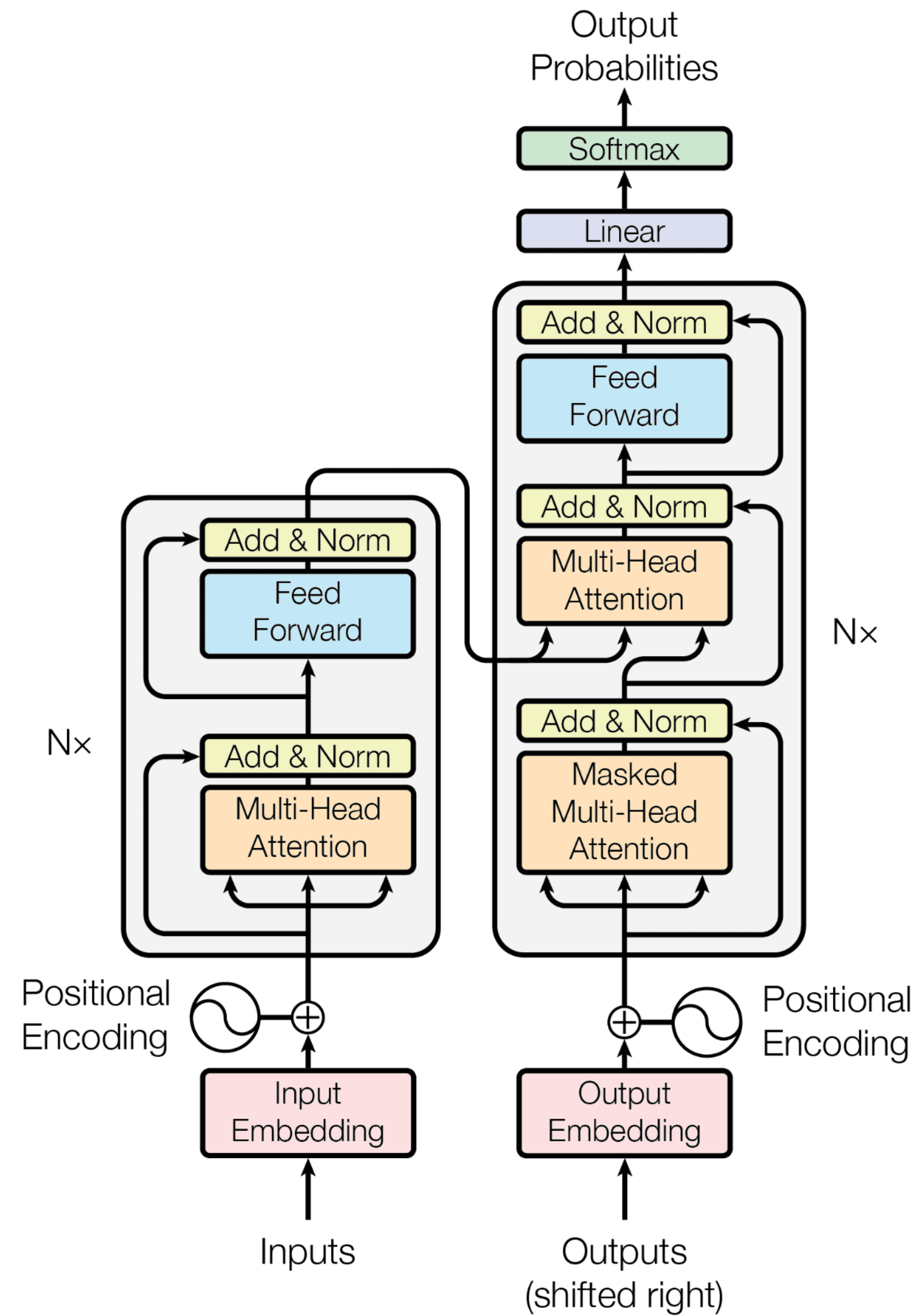
Attention



Video



Transformer



Pre-training

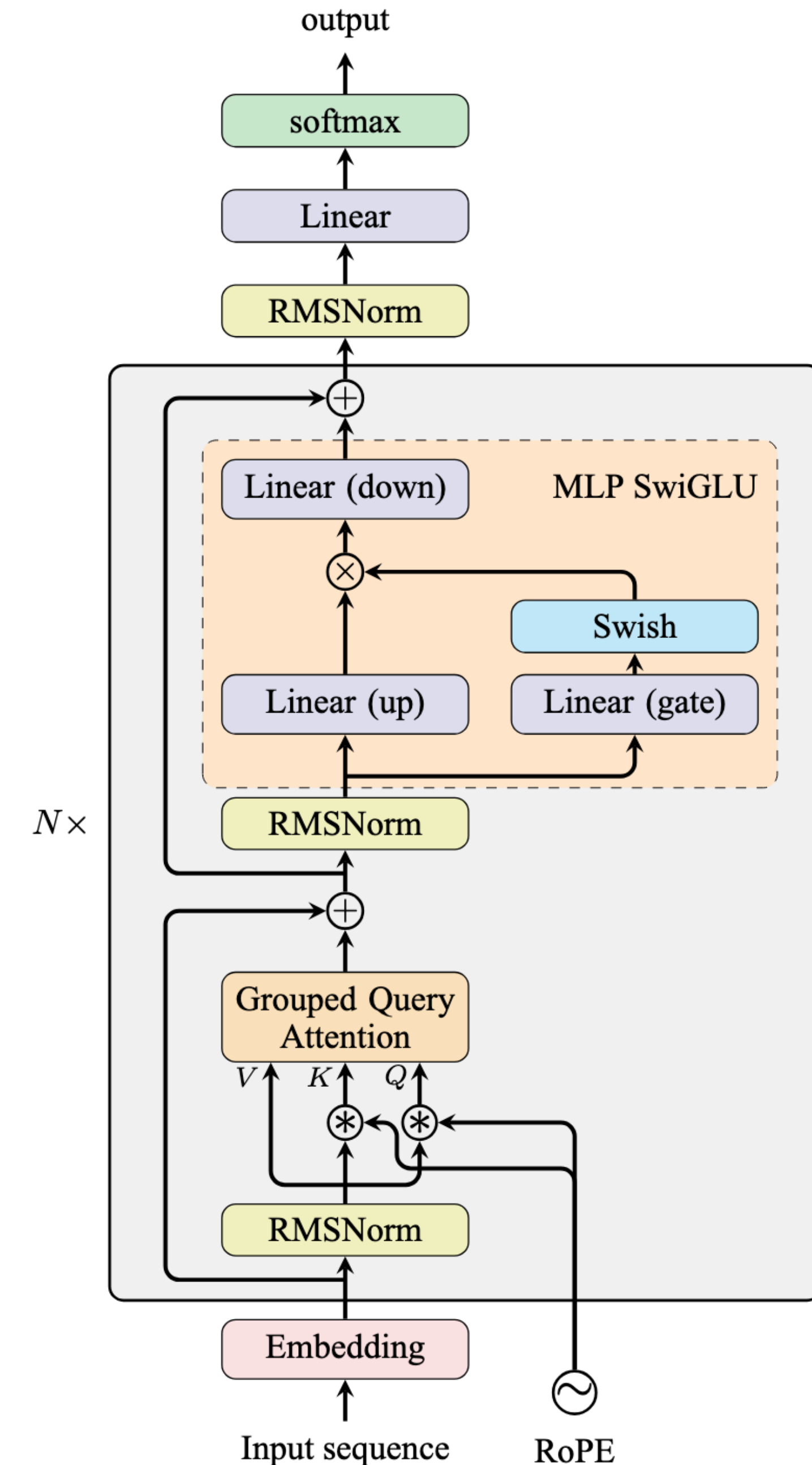
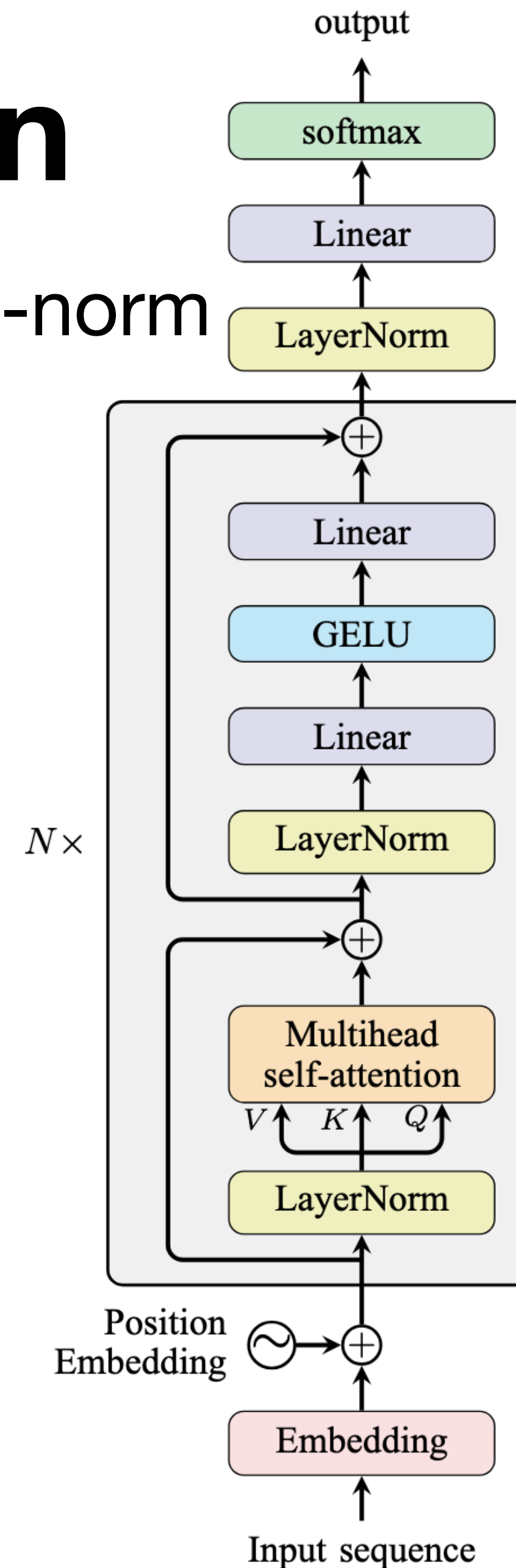
SFT

RLHF

$$\text{Loss} = -\log(\text{policy_selected}) + \log(\text{policy_rejected}) + \text{KL}(\text{policy}(\text{prompt}))$$

Qwen Model Innovation

GQA, SwiGLU, RoPE, RMSNorm with pre-norm



Nega GQA (Grouped Query Attention)?

- To cut memory cost of KV Cache

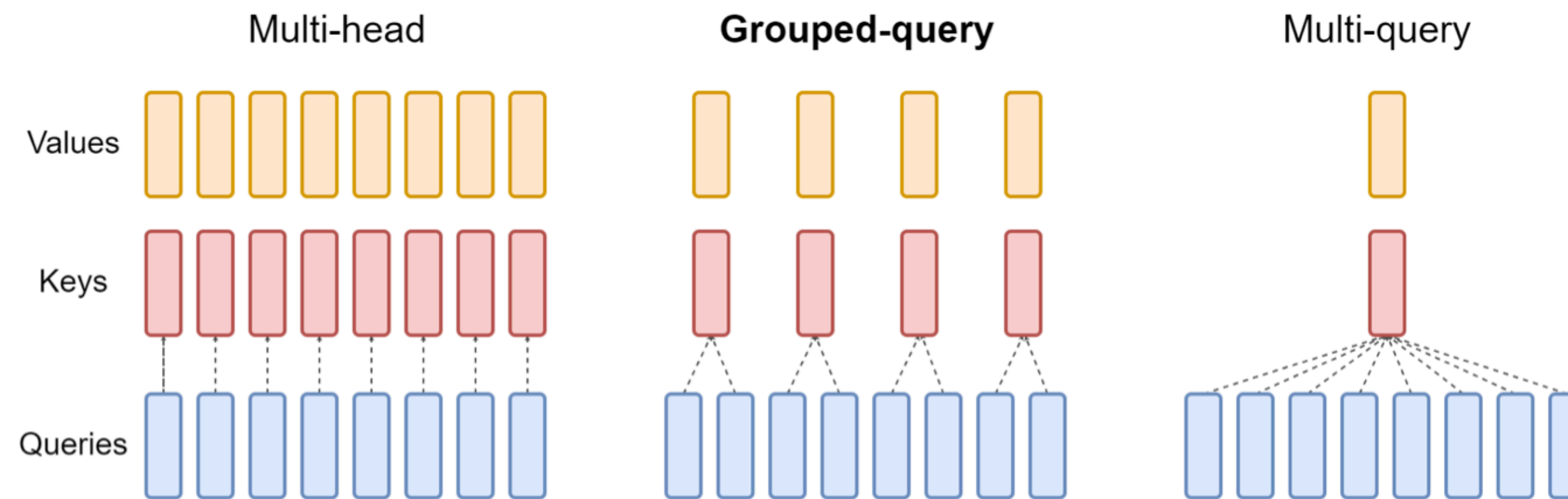


Figure 2: Overview of grouped-query method. Multi-head attention has H query, key, and value heads. Multi-query attention shares single key and value heads across all query heads. Grouped-query attention instead shares single key and value heads for each *group* of query heads, interpolating between multi-head and multi-query attention.

Nega SwiGLU (Swish-Gated Linear Unit)? Efficient Data Flow

$$\text{SwiGLU}(x, W, V, b, c, \beta) = \text{Swish}_{\beta}(xW + b) \otimes (xV + c)$$

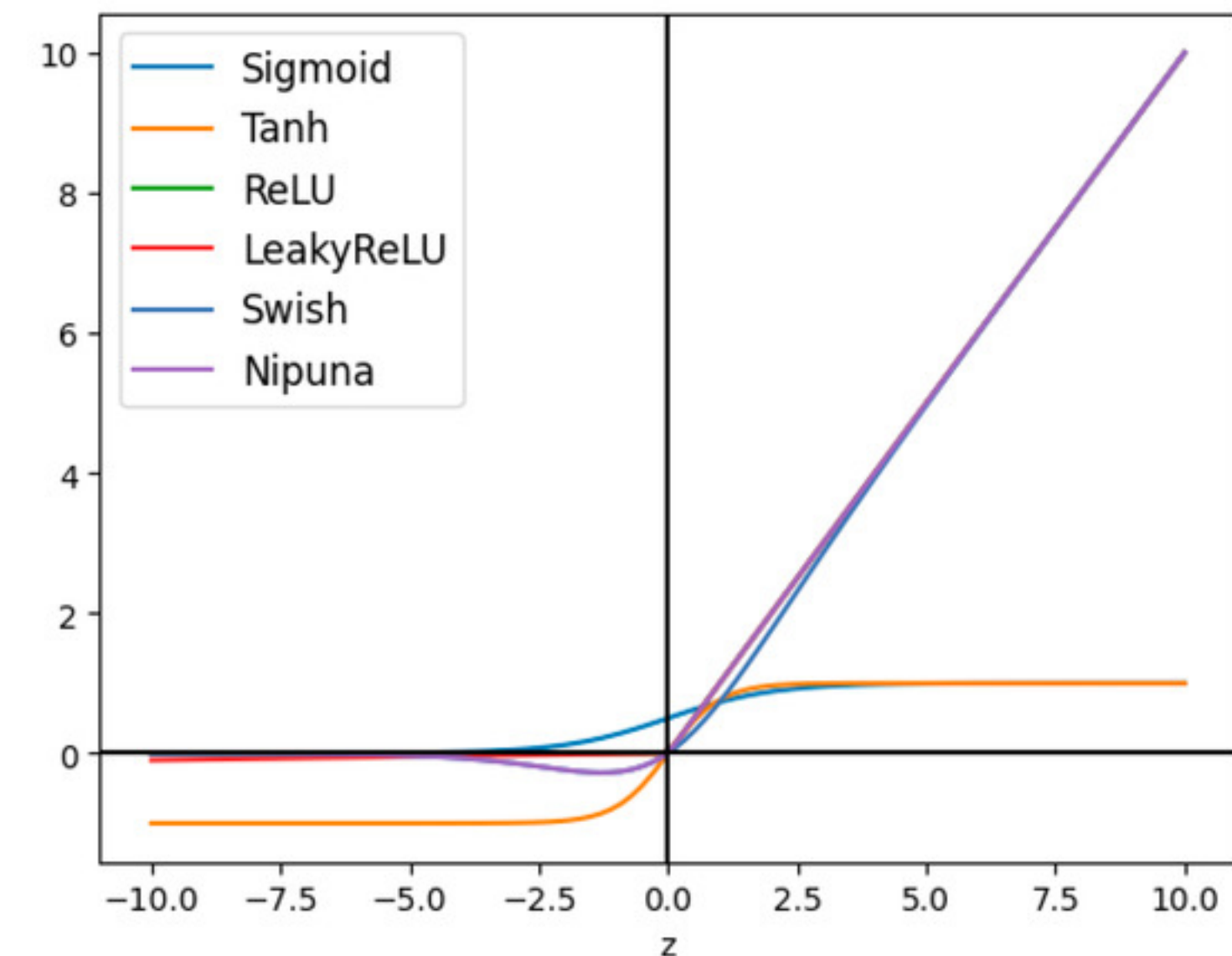
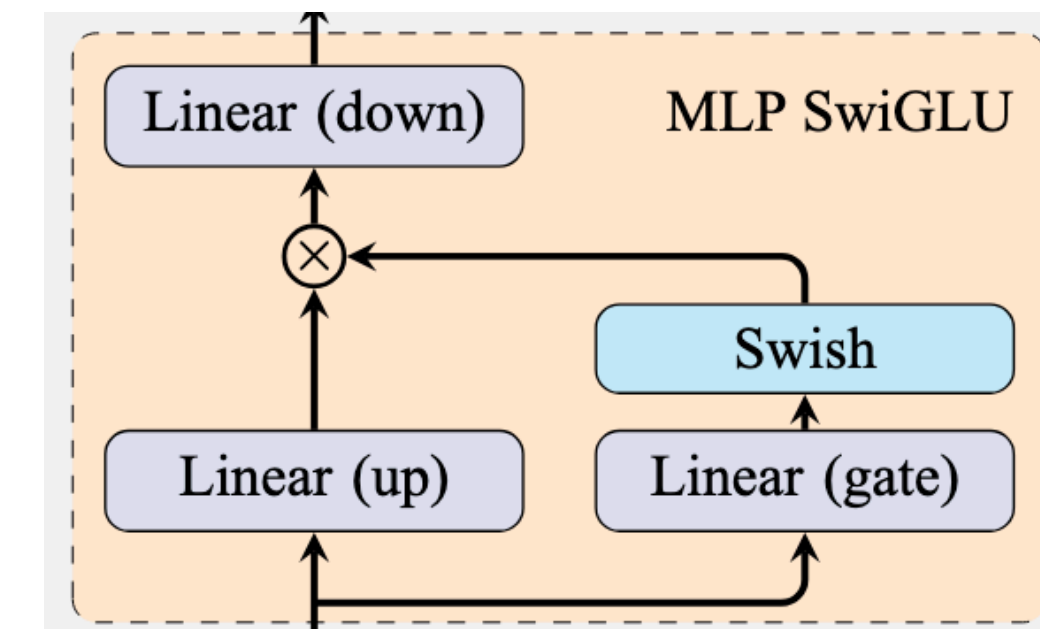
W - learnable matrix (1024, 3072)

V - learnable matrix (1024, 3072)

W_2 (Linear down) - (3072, 1024)

c, b - biases

β - [0, 1] hyperparam



Nega RoPE?

- More intuitive
- Relative, not absolute
- Rotation happens only in several embedding dims.

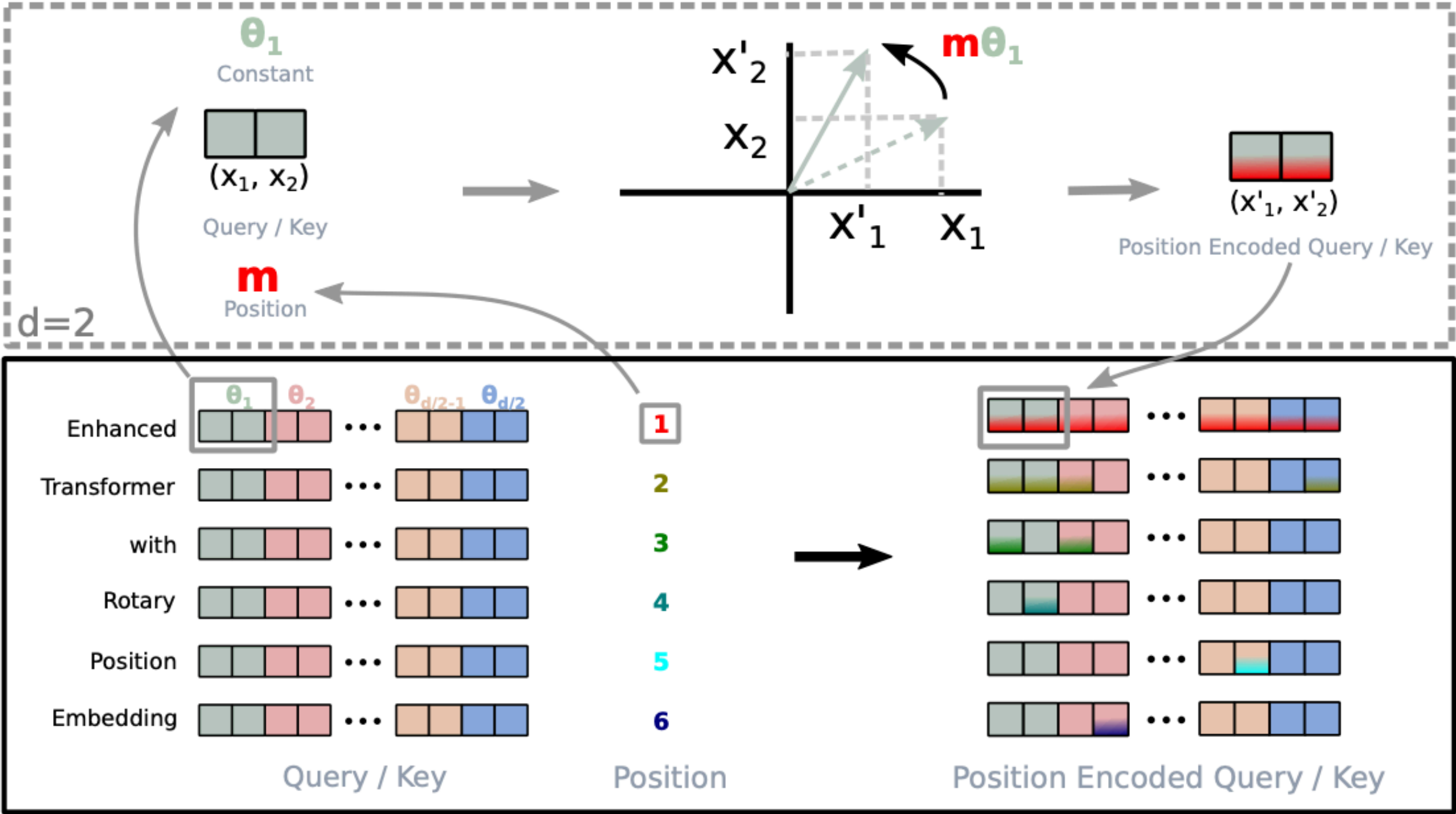
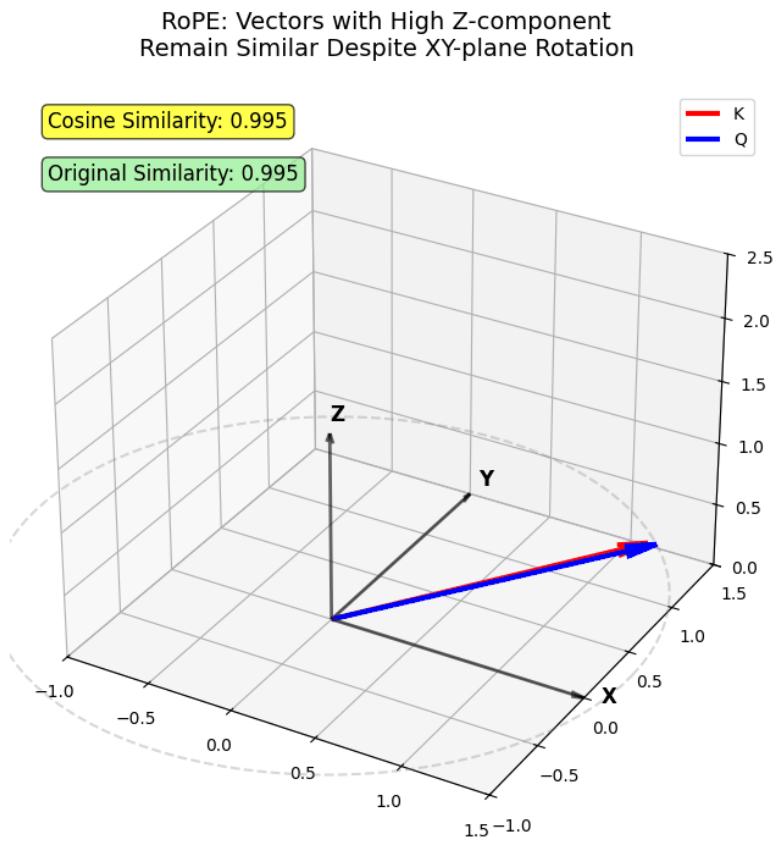
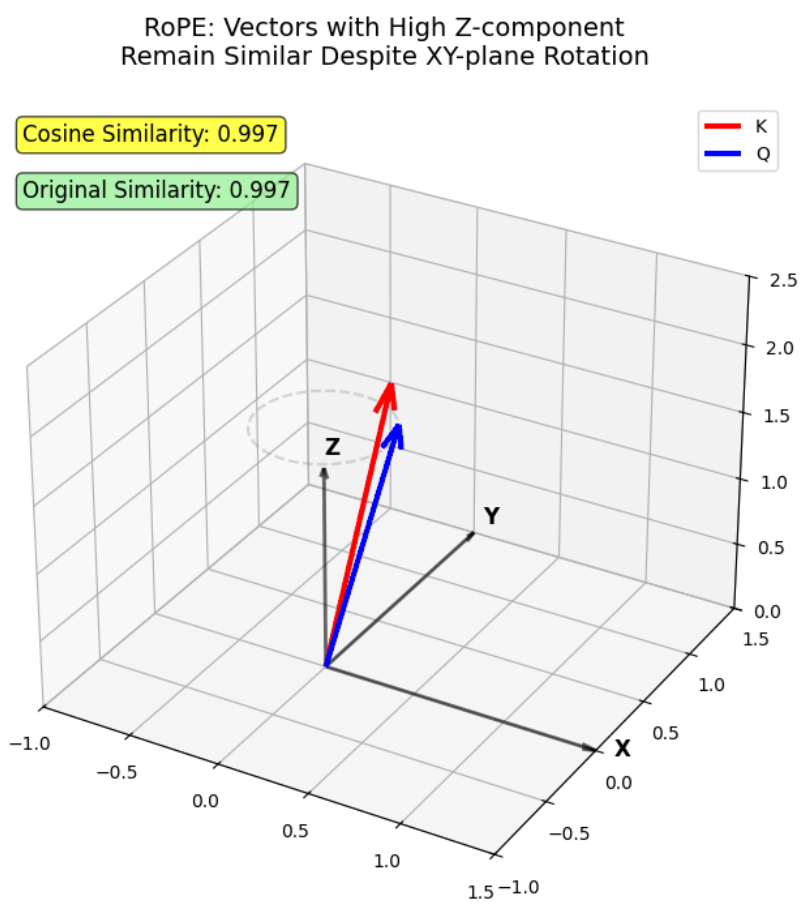
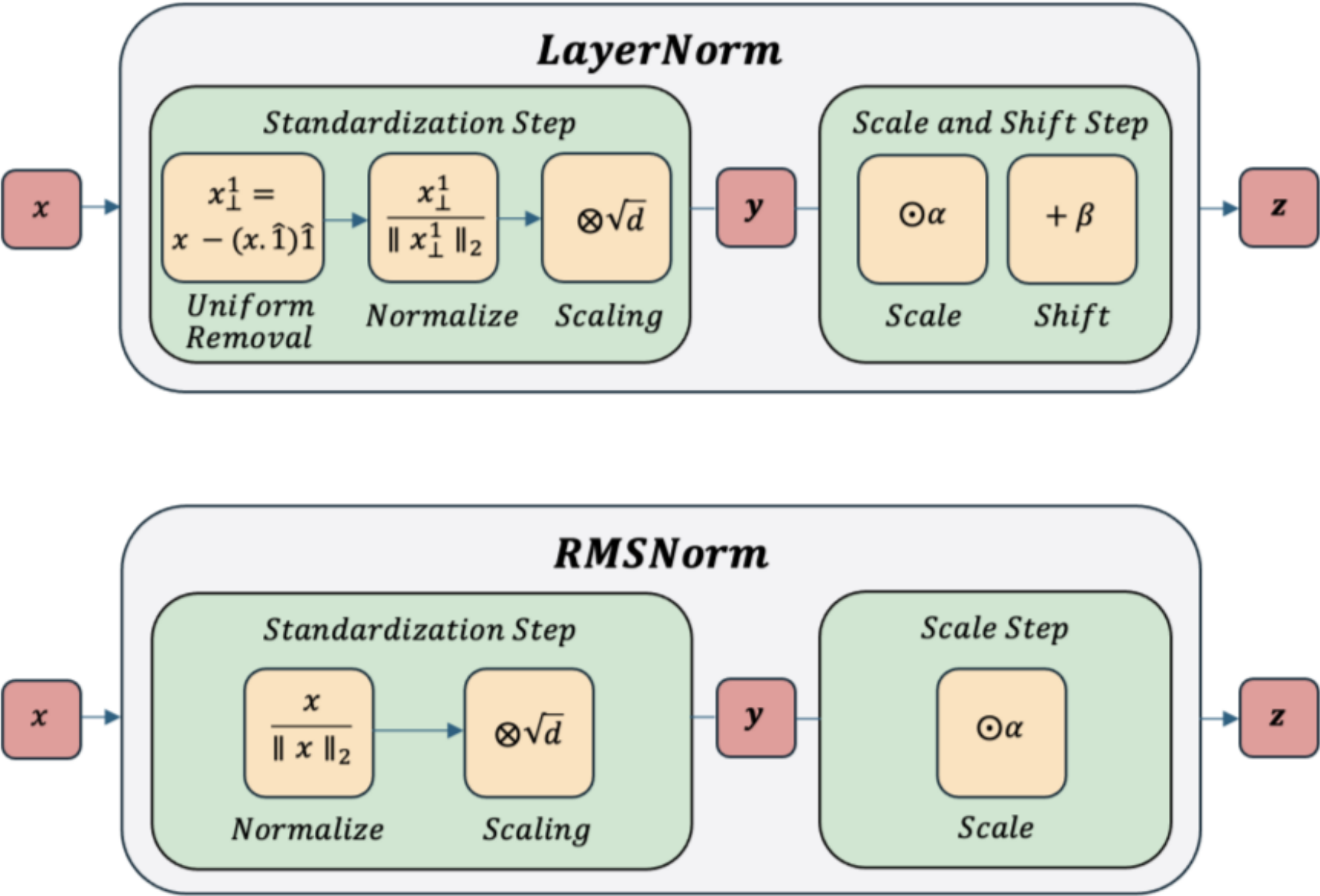
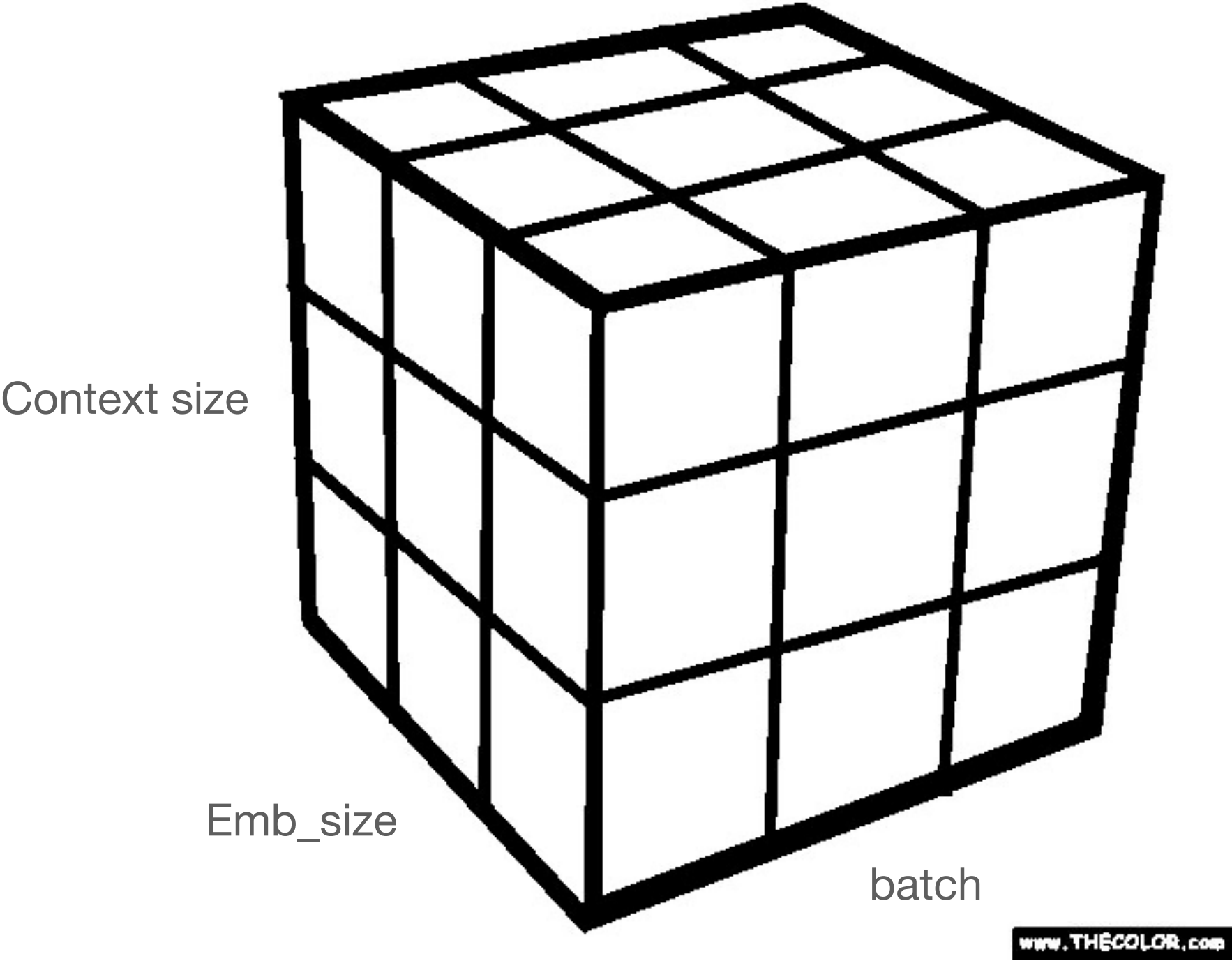


Figure 1: Implementation of Rotary Position Embedding(RoPE).



Nega RMSNorm

- Simplified LayerNorm (easier)



	Weight matrix re-scaling	Weight matrix re-centering	Weight vector re-scaling	Dataset re-scaling	Dataset re-centering	Single training case re-scaling
BatchNorm	✓	✗	✓	✓	✓	✗
WeightNorm	✓	✗	✓	✗	✗	✗
LayerNorm	✓	✓	✗	✓	✗	✓
RMSNorm	✓	✗	✗	✓	✗	✓
pRMSNorm	✓	✗	✗	✓	✗	✓

Table 1: Invariance properties of different normalization methods. “✓” indicates invariant, while “✗” denotes the opposite.

Qwen1.0

Qwen2.0

Qwen2.5

Qwen3.0

Qwen3 (reasoning + architectural cleanup)

Innovations: unified thinking and non-thinking modes in one switchable model (no more separate chat vs reasoning model);

adds QK-Norm for training stability;

Qwen3-MoE drops the shared experts Qwen2.5-MoE used and adopts a global-batch load-balancing loss for expert specialization;

multilingual support jumps from 29 to 119 languages.

Qwen3.0

3.2 Pre-training Stage

The Qwen3 models are pre-trained through a three-stage process:

- (1) **General Stage (S1):** At the first pre-training stage, all Qwen3 models are trained on over 30 trillion tokens using a sequence length of 4,096 tokens. At this stage, the models have been fully pre-trained on language proficiency and general world knowledge, with training data covering 119 languages and dialects.
- (2) **Reasoning Stage (S2):** To further improve the reasoning ability, we optimize the pre-training corpus of this stage by increasing the proportion of STEM, coding, reasoning, and synthetic data. The models are further pre-trained with about 5T higher-quality tokens at a sequence length of 4,096 tokens. We also accelerate the learning rate decay during this stage.
- (3) **Long Context Stage:** In the final pre-training stage, we collect high-quality long context corpora to extend the context length of Qwen3 models. All models are pre-trained on hundreds of billions of tokens with a sequence length of 32,768 tokens. The long context corpus includes 75% of text between 16,384 to 32,768 tokens in length, and 25% of text between 4,096 to 16,384 in length. Following Qwen2.5 (Yang et al., 2024b), we increase the base frequency of RoPE from 10,000 to 1,000,000 using the ABF technique (Xiong et al., 2023). Meanwhile, we introduce YARN (Peng et al., 2023) and Dual Chunk Attention (DCA, An et al., 2024) to achieve a four-fold increase in sequence length capacity during inference.

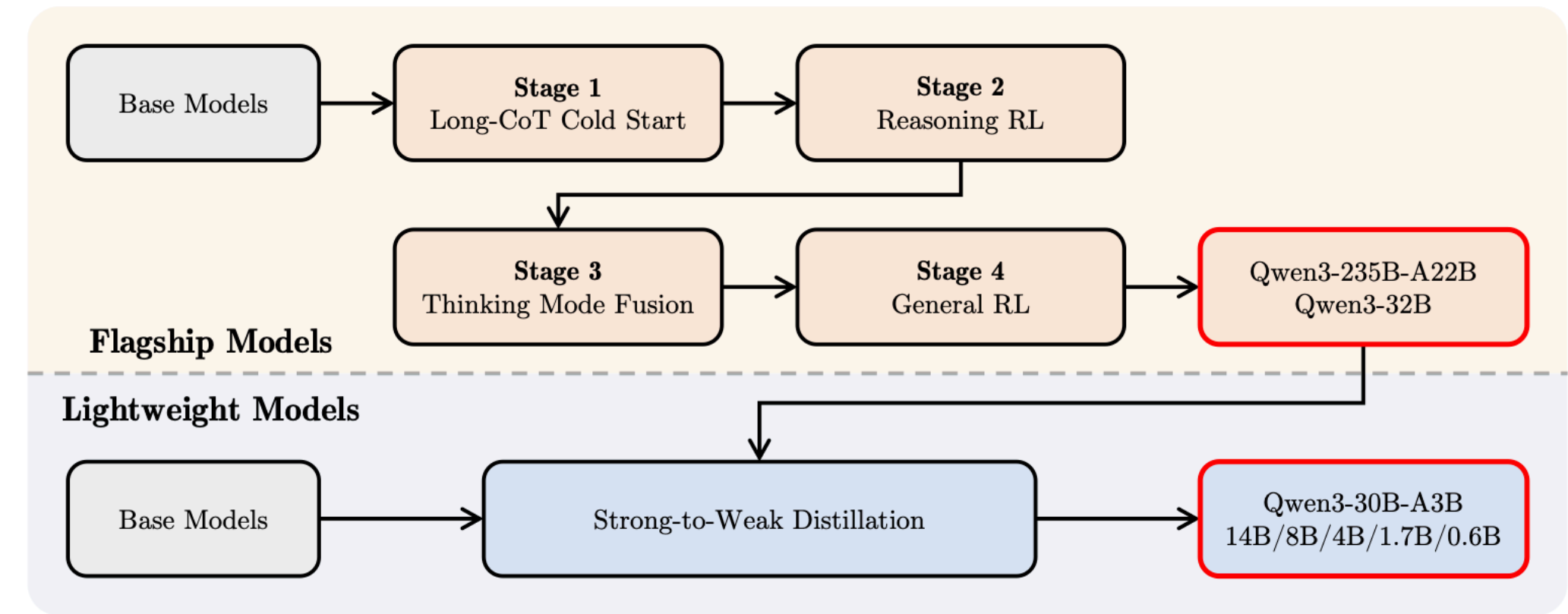


Figure 1: Post-training pipeline of the Qwen3 series models.

Table 1: Model architecture of Qwen3 dense models.

Models	Layers	Heads (Q / KV)	Tie Embedding	Context Length
Qwen3-0.6B	28	16 / 8	Yes	32K
Qwen3-1.7B	28	16 / 8	Yes	32K
Qwen3-4B	36	32 / 8	Yes	128K
Qwen3-8B	36	32 / 8	No	128K
Qwen3-14B	40	40 / 8	No	128K
Qwen3-32B	64	64 / 8	No	128K

Table 2: Model architecture of Qwen3 MoE models.

Models	Layers	Heads (Q / KV)	# Experts (Total / Activated)	Context Length
Qwen3-30B-A3B	48	32 / 4	128 / 8	128K
Qwen3-235B-A22B	94	64 / 4	128 / 8	128K

Table 9: **Examples of SFT data for thinking and non-thinking modes during the thinking mode fusion stage.** For the thinking mode, the /think flag can be omitted since it represents the default behavior. This feature has been implemented in the chat template¹ supported by the Hugging Face’s tokenizer, where the thinking mode can be disabled using an additional parameter enable_thinking=False.

Thinking Mode	Non-Thinking Mode
< im_start >user {query} /think< im_end > < im_start >assistant <think> {thinking_content} </think> {response}< im_end >	< im_start >user {query} /no_think< im_end > < im_start >assistant <think> </think> {response}< im_end >

Google Collab Mashq

Colab

Tokenizer, MoE, Dense layers, actual output

O'zbek LLM

Nega kerak?



Kun.uz
https://kun.uz › news › 2025/08/06 › uzbekistan-to-dev... ⋮

Uzbekistan to develop national AI language model ...

Aug 6, 2025 — **Uzbekistan to develop national AI language model** to preserve cultural identity and ensure digital sovereignty. [Read more](#)

Class	Description	Representative Datasets	Primary Use in Model Training
Raw Text Corpora	Unstructured collections of text (books, news, and web pages).	<i>Tahrirchi/uz-crawl, murodbek/uz-books</i>	Pretraining: builds fundamental Uzbek grammar, syntax, and style.
Instruction Datasets	Prompt-response examples showing how to follow user instructions.	<i>UAzimov/uzbek-instruct-llm, Behbudiy/alpaca-cleaned-uz, Behbudiy/translation-instruction</i>	Supervised fine-tuning: teaches task execution and conversational behavior.
Chain-of-Thought (CoT) Data	Multi-step reasoning traces showing how conclusions are reached.	<i>(No publicly available Uzbek dataset yet)</i>	Reasoning fine-tuning: trains logical, interpretable problem solving.
Annotated / Task-Specific Data	Labeled datasets for NLP tasks (e.g., sentiment, NER, semantic similarity).	<i>Behbudiy/uzbek-sentiment-analysis NER for Uzbek (Mendeley)</i>	Evaluation and specialization: supports classification, labeling, or entity recognition.
Evaluation datasets	Benchmarks used to measure model performance in language understanding, reasoning, and instruction following	<i>Tahrirchi/uzlib, SimRelUz</i>	<i>Performance assessment</i> — evaluates comprehension, factual accuracy, fluency, across diverse Uzbek language tasks.

Savollar va Javoblar

Extra:

<https://colab.research.google.com/drive/1GSYSJkckcFF0vfFPnGC7wr9sUrrfUPRu>

<https://colab.research.google.com/drive/1LWY3tW9B2x7dlEOqr-b81SgXME9P1SnY>

<https://colab.research.google.com/drive/1eulXqgajSkFlz9l0itw7aL1SmH3664LP#scrollTo=QyEjW-WuYQIm>

Feedback QR Code and Link: (1 daqiqa holos)

